Lesson 6: Victory Condition

Godot Engine

VIDEO GAME CREATION

This week: a real win condition. Your job is to track how many enemies are left and show "You Win!" the moment the last one is destroyed - completing your game's Level 1 MVP milestone.

Open your own [YourName]Game project from Lesson 5 - don't start a new one. Everything this week builds on the row of enemies you already have.

Tip: no new assets this week - "You Win!" is shown with each engine's own basic text tools, not an image.

Setup

1. Open your [YourName]Game project from Lesson 5.
2. Add a Label node to your main scene, rename it WinText, set its Text to "You Win!", and hide it (uncheck Visible in the Inspector, or call hide() in code).
3. Add a new Node as a child of your main scene, rename it GameManager, and attach a new script to it - this is the one place win-tracking will live.
4. In enemy.tscn's own script, add a new signal at the top: signal destroyed.

Level 1 - Minimal (MVP)

The moment the last enemy is destroyed, "You Win!" appears on screen. This completes your game's Level 1 MVP milestone - a complete, playable core loop.

5. In enemy.tscn's _on_hitbox_area_entered(), call destroyed.emit() right before queue_free() (code below).
6. In GameManager's script, add a var enemies_remaining and a function on_enemy_destroyed() that decrements it and shows WinText at zero.
7. In the main scene's script (which still does the spawning, same as Lesson 5), connect each new enemy's destroyed signal to $GameManager.on_enemy_destroyed instead of a method on the main scene itself.
8. Run the scene and destroy every enemy to test - "You Win!" should appear the instant the last one is destroyed.

```
# enemy.tscn's script - add at the top and inside the hit handler:
signal destroyed

func _on_hitbox_area_entered(area):
    area.get_parent().queue_free()
```

```
    destroyed.emit()
    queue_free()

# GameManager's script:
var enemies_remaining := 5

func on_enemy_destroyed():
    enemies_remaining -= 1
    if enemies_remaining <= 0:
        get_node("../WinText").show()

# Main scene's script:
func _ready():
    for enemy_count in range(5):
        var new_enemy := enemy_scene.instantiate()
        new_enemy.position = Vector2(40 + enemy_count * 60, 40)
        new_enemy.destroyed.connect($GameManager.on_enemy_destroyed)
        add_child(new_enemy)
```

Level 2 - Complete (Alpha/Beta)

The win text appears exactly once, exactly when the last enemy is destroyed - not early, not late, and not repeatedly.

- Destroy all 5 enemies and confirm "You Win!" appears exactly once, exactly when the last one is destroyed - not one enemy early, and not repeatedly.
- This already works correctly because destroyed.emit() fires exactly once per enemy (right where it's actually destroyed), and each enemy's signal is connected exactly once, always to GameManager's on_enemy_destroyed() - there's no way for the count to drift.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: add an AudioStreamPlayer node (not 2D, since it's UI, not positional) as a child of GameManager, assign the provided sfx_win.wav as its stream, and call it from GameManager's own script right where WinText is shown.
- Visual: instead of just show()-ing WinText, animate it in - a quick scale tween using create_tween() reads more like a celebration than a plain appearance.
- Structure: keep the win-tracking (enemies_remaining, the check for zero) entirely inside GameManager's own script - the main scene only ever spawns enemies and wires up their signals, and enemy.tscn only ever announces that it was destroyed. Neither one tracks or decides anything itself.

If Something's Wrong

- "You Win!" never appears: confirm destroyed.emit() is actually being called, and that new_enemy.destroyed.connect($GameManager.on_enemy_destroyed) runs for every spawned enemy, not just some of them.
- "You Win!" appears too early: confirm enemies_remaining's starting value on GameManager actually matches the number of enemies your loop spawns.
- Error about a missing signal: confirm signal destroyed is declared at the very top of enemy.tscn's script, outside any function.

Want More?

Student Resource List > Godot section, or docs.godotengine.org